

Vision Empower Trust AI Skilling Program

Introducing Artificial Intelligence, Responsible, Informed, Conscious (RIC) Use of AI MODULE-1

Vision Empower Trust



Agenda

Item	Details
Audience	Teachers, special educators, resource persons and school leaders working with visually impaired students
Duration	90 minutes
Mode	Facilitator-led, discussion-based, with one short hands-on or scenario activity
Prerequisites	Basic smartphone/computer familiarity; no prior AI knowledge required
Module purpose	To introduce AI formally, explain how it differs from earlier technologies, clarify AI/ML/Generative AI, show key capabilities, promote Responsible, Informed and Conscious AI Use
Session principle	Teacher judgement remains central. AI is an assistant for access, planning and communication; it is not a replacement for the teacher.

Pre-Assessment

- Do you know the difference between Traditional technologies vs AI technologies?
- Do you know the difference between supervised and unsupervised learning ?
- What is the role of data in making an AI solution useful?
- What does human-in-the-loop mean and why is it important?
- Can AI make mistakes ?
- What is hallucination ?
- Does AI have emotions ?
- Do you think Responsible, Informed and Conscious use of AI is necessary?

Ice Breaker – Thinking about different technologies

Objective

- Recall technologies and the problems they solve.
- Reflect on the following:
 - What problem did it solve?
 - How does it help today?

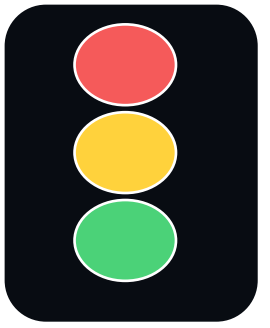
Key Takeaway

- Technology has evolved from the Abacus to AI, making life, learning, and accessibility easier.
- Some technologies are not present anymore, they have evolved and other tools have taken their place

Ice Breaker – Can you spot the AI ?

- Do you use AI ? How ?
- Artificial intelligence is used in a number of ways in our daily lives. Can you spot the AI?
 - Traffic light changing at fixed intervals
 - Face Recognition System
 - Washing machine with preset timer
 - Voice assistant
 - YouTube video recommendations
 - Spell check & grammar tools
 - Google Maps route suggestions
- Some technologies are **automation based**, but most are **evolving with AI integration**

Traffic lights – Without and With AI



LEVEL 1 — MANUAL / FIXED

- Signal changes after fixed time
- Example: Green = 60 seconds
- Same schedule whether road is empty or crowded
- Simple, predictable, but inefficient

LEVEL 2 — RULE BASED AUTOMATION

- Sensors detect traffic
- Rules decide the next signal
- Example: extend green if more vehicles are waiting
- Faster response, but rules must be predefined

LEVEL 3 — DATA DRIVEN AI

- Uses current + historical traffic data
- Predicts congestion
- Optimizes signal timing across intersections
- Can improve its decisions from data

Understanding AI from basics

What is Intelligence ?

- Is it only to know? No, it is the **ability to learn, think, understand, solve problems, and apply knowledge** effectively
 - Not just knowing, but a **combination** of many mental abilities
 - Ability to adapt to new situations and make appropriate decisions based on **experience**
 - Ability to work with **gaps in information, to estimate, to guess**
-
- Do you agree? Do you have any other inputs?

What is Artificial Intelligence ?

- Artificial Intelligence (AI) is the field of computer science that **enables machines to perform tasks that tries to mimic human intelligence**.
- These machines can conduct **complex thinking processes** such as data analysis, pattern recognition, prediction of trends, solving problems, and decision-making by using data
- AI systems are **designed to learn, adapt, and improve their performance** over a period of time (experience through data).
- **AI = Man-made, Computing + Intelligence**
- Artificial Intelligence **cannot be equated** to Human Intelligence which includes more aspects like empathy, conscience and ethics
- It is acquired **through data and training** by humans
- Many advancements in Science, Technology, Engineering and Maths have led to the development of AI

Why is AI important?

- Some problems **cannot be solved with traditional programming** – facial recognition, handwriting recognition
- AI **can learn from huge amounts of data** (millions of records, data, images, signals etc.)
- AI **can find patterns** across very large datasets.
- AI **can automate** complex cognitive tasks
- AI **can help humans** make better decisions
 - Predicting traffic congestion
 - Forecasting weather
 - Detecting financial fraud
- AI **can make technology more accessible**
 - Describe images , Convert speech to text , Convert text to speech , Recognize objects , Translate languages , Generate alternative descriptions
- AI **can apply learnings to create** new solutions and data

Where is AI being used ?

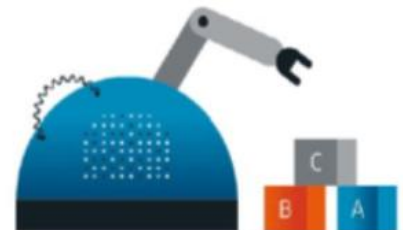
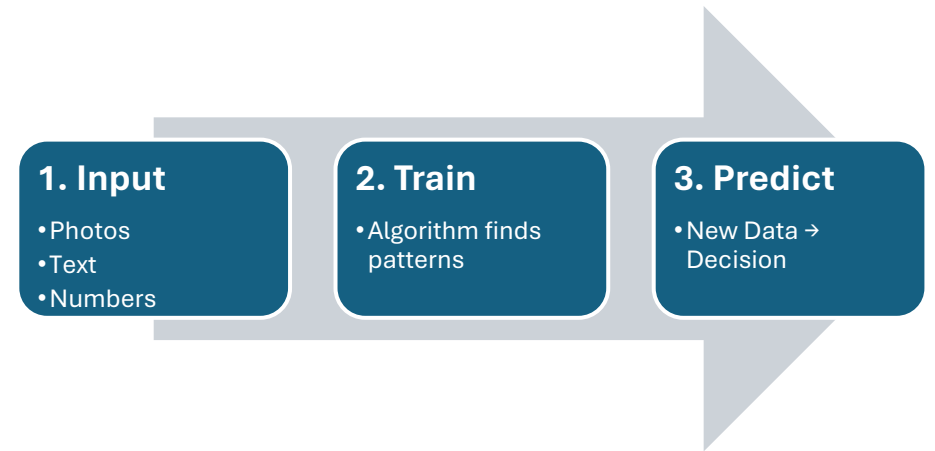
- Medical Applications
 - Kidney stone detection
 - Brain Tumor detection
- Office automation
- Banking applications
- Agricultural applications
- Weather forecasting
- Traffic management (smart signals)
- Education
- Almost all the fields

- Do you know any other fields?

How to make a machine
intelligent?

How to make a machine learn and think?

- Instead of explicitly programming rules ("If X do Y"), we feed the computer data and let it learn the rules itself.
- Traditional Programming
 - Program → Computer → Output
 - "If X happens → do Y."
- Machine Learning
 - Data + Output → Computer → Program (Model)
 - Training the Model with Data
 - "Here are thousands of examples → learn the pattern."
- AI is not new; it has been evolving since 1950s from foundations, to internet era, deep learning and generative AI



Machines can be taught in many ways

- Like a child learning with a Teacher – **Supervised Learning**
 - A teacher showing a child flashcards/ objects. “This is an Apple”, “This is a Banana”. Later, the child identifies them alone.
 - Learning with the help of trainer provided input and their labels. Once trained, AI can **predict**.
- Like a child learning on its own – **Unsupervised Learning**
 - Giving a child a set of buttons of various shapes, sizes and colours and ask them to sort on their own
 - Once the groups are created, meaning is added. Once trained, AI can **classify**.
- Like training a dog to sit with treat/no treat – **Reinforcement Learning**
 - Learning by trial and error; Award for right answer and penalty for wrong answer
 - Over many experiences, patterns are learnt. Once trained, AI can **perform desirable actions**

Machines can be taught in many ways (continued)

- Learning complicated patterns with many layers of analysis – **Deep Learning**
 - Like human brain, inspect, store, identify, mix, match features of various types
 - Used for image recognition, natural language processing, computer vision etc. Once trained, **AI can recognize.**
- Creating new content based on what was learnt from the earlier data in response to a “prompt” from the user - **Generative AI**
 - Content can be text, images, audio, video, and code
 - Generation based on what has the highest probability to be valid and applicable in the context, within the limits of time and storage
 - Can be erroneous without any strict bounds for adhering to sources / limits on the creativity
 - Once trained, **AI can create.**

Lets experience AI

Shall we use AI and experience it?

- Generating alternate text for VI students to understand pictures – Using Seeing AI
- Create interesting audio clips - Suno <https://suno.com/>
- Asking examples to make a hard concept easily understood
 - “Generate some interesting and intuitive examples for a VI child in TamilNadu to understand about the concept of Momentum clearly”
- Do you want to share how you use AI?

Data – The Fuel of AI

Data – The fuel of Artificial Intelligence

- Data is **the fuel and source of intelligence** for AI Models
- Data means **raw or processed inputs** that can be further processed and analysed to get useful information which provides meaningful insights for decision-making.
- Data collected over a period of time shows **patterns** – contributes to experience and knowledge
- Can be used to take decisions and predict or forecast
- Collection of data is called a **“Dataset”**
 - Huge efforts to create comprehensive datasets are undertaken to make the AI systems better
 - For example,
 - Dataset of cough sounds mapped to whether COVID was found in a patient with such a cough
 - Dataset of bird sounds to identify the birds
 - Dataset of human faces for facial recognition
 - Dataset of movies watched by users

How to make data-driven intelligence systems better?

- You might have faced the issue that speech recognition works well with English compared to other regional languages – Why is that the case?
- Sometimes, the responses we get may not be relevant for our region/ culture – Why is that the case?
- Facial recognition software was not able to recognize facial images from all countries, ethnicities etc. Why did that happen?
- Depends on the **data** used for building the models
- Do humans use data?
 - Yes, just like AI, we need **good data** to take proper decisions as well
 - However, **we do not use data alone**. We **also use our intelligence, conscience and intuition**.
 - In the absence of these in a machine, **only data contributes majorly to artificial intelligence** and that can lead to mistakes and errors

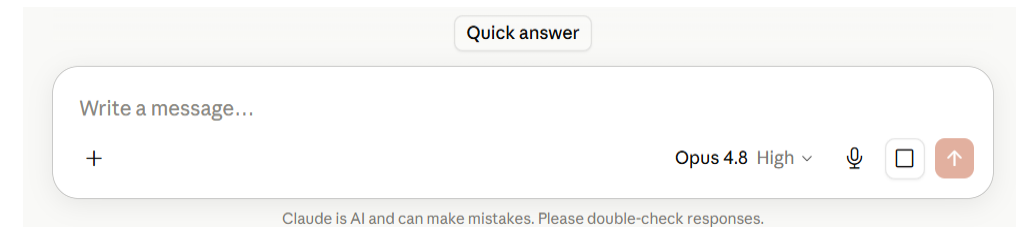
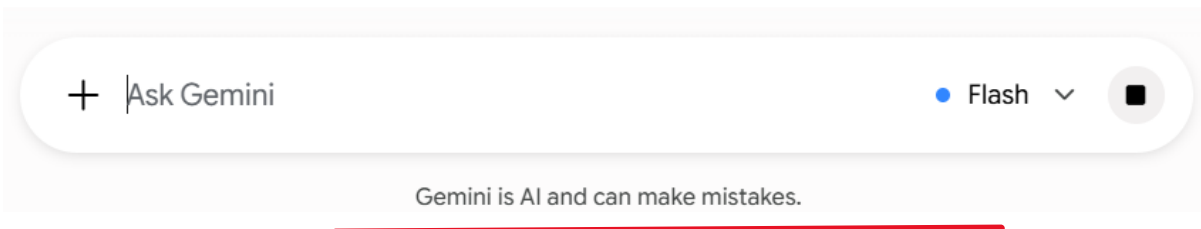
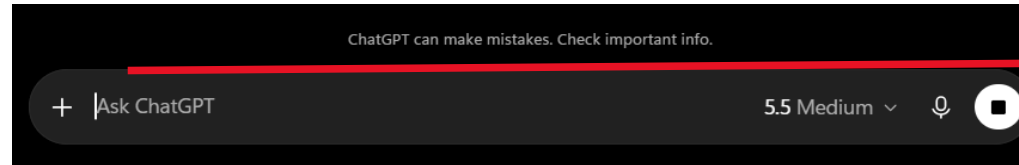
Good Data and Design – Needed for AI

- How to minimize these errors from data-driven intelligence? The data used for training should be
 - Very large to include as many variations as possible
 - Representative to not include bias for any particular set
 - Good quality for the machine to be able to accurately understand all features
- If the quality of data is bad, then the decisions taken by the AI model will also be inaccurate
- Any bias in data translates into inaccuracies, bias, inequalities, incorrect predictions etc.

Towards Responsible, Informed
and Conscious use of AI

AI is important and useful. Why should we be concerned ?

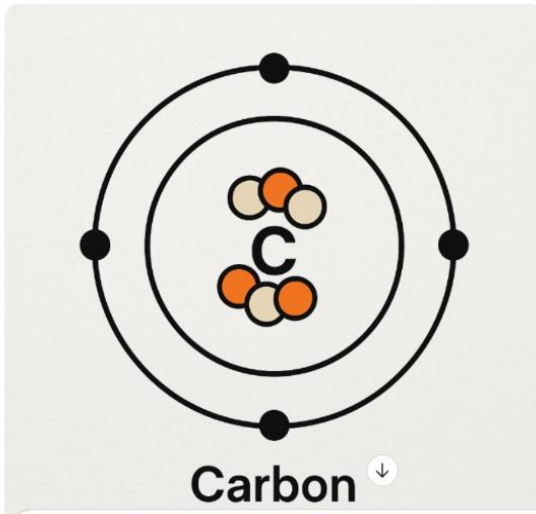
- Why do all GenAI platforms say that they can make mistakes?
 - Was this warning included in a calculator? Why not?
 - Have you experienced incorrect responses from any AI tool?
 - Have you seen that it has generated wrong content and says “Sorry”, “you are right” etc.
 - Have you heard about excessive AI dependence resulting in extremes like illnesses and fatalities
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- Why do you think this is happening?
 - What does all this mean to the users? Can you trust the AI output completely?



Prompt : “Draw the structure of the carbon atom”

draw the structure of a Carbon atom

Image created



A carbon atom has 6 electrons.

Here's the breakdown:

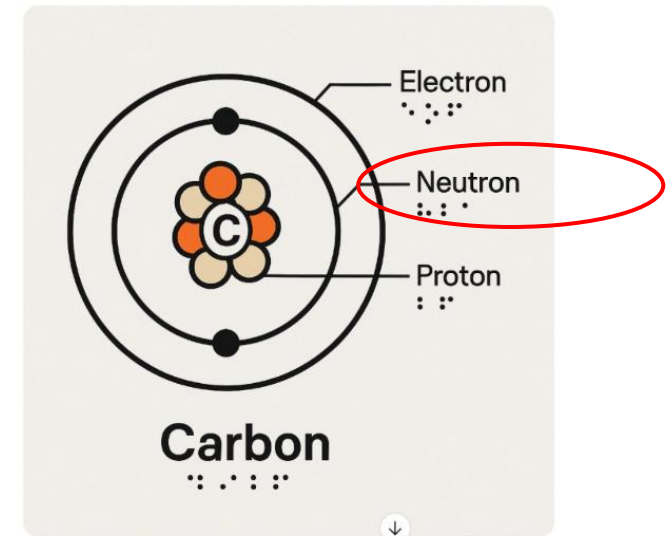
- Atomic number of carbon = 6 → this means it has 6 protons and 6 electrons (in a neutral atom).
- Electron arrangement:
 - K shell (1st shell): 2 electrons
 - L shell (2nd shell): 4 electrons

So, the electron configuration of carbon is 2, 4.



how many electrons are there in a carbon atom

Image created



but in the picture generated above, 6 electrons were not there. It is inaccurate. How was it generated.

Thought for 1m 4s

You're right—and I'm sorry about that.

That picture was created by an AI image generator from a text prompt I wrote. It's not a physics/CAD tool, so even with instructions it can miscount or misplace elements (e.g., fewer than 6 electrons, labels not exactly as specified). No manual drawing was done.

Understanding GenAI Responses

- Responses depend on the **quality of data used for training and the design of the system**
- **Inherent design** of the AI models can add to inaccuracies and hallucination
- Generation with AI is **not always backed with actual facts**. User needs to instruct and bound the creation with context and expectations of accuracy
- How the machine learns is not completely explainable
- Responsibility of **writing a good “prompt”** is with the user, better prompt may lead to better results
- Users need to be able to **assert the accuracy** of the content “created”
- Most Gen-AI solutions are from private corporations and the free plans may not be high performing in terms of quality

How do we use AI if it has pros and cons?

- **Humans are accountable** for what we use AI for and how we use its outputs
- We cannot penalize AI for its mistakes, inaccuracies and errors
- **Responsible, Informed and Conscious (RIC) Use** of AI needs to be practiced
- **Responsible :**
 - Use AI in a manner that does not result in harm for yourself or others
- **Informed :**
 - Know that AI can make mistakes, can have bias, can make mistakes, does not have emotions
 - Know that AI can generate content without it being reality (hallucination)
- **Conscious :**
 - Use AI with full understanding of the accountability for actions based on the information or suggestions that are got from the AI solutions
- An **Responsible, Informed and Conscious** human in the loop is the best way to use AI

AI In Education

Covered in detail in Module 2

Examples of data from education

Data Type	Examples in School Education	How it can be used
Numerical Data	Student marks, attendance percentage, number of assignments submitted, test scores, class strength, time spent on an LMS	Performance analysis, identifying learning gaps, tracking progress
Text Data	Essays, answers to descriptive questions, teacher feedback, lesson plans, report cards, discussion forum posts, textbook content	Assess writing skills, generate summaries, analyze feedback, create learning resources
Image Data	Scanned answer sheets, diagrams drawn by students, photographs of science experiments, artwork, geometry figures, handwritten notes	Automated evaluation, handwriting recognition, diagram analysis, portfolio assessment
Audio Data	Students reading aloud, language pronunciation exercises, oral exams, teacher lectures, classroom discussions, podcast lessons	Speech assessment, pronunciation feedback, transcription, accessibility
Video Data	Recorded classroom lessons, student presentations, lab experiment demonstrations, sports activities, online classes, classroom observations	Teacher training, student assessment, engagement analysis, flipped learning

Where can AI be used in Education

- Lesson plan generation
- Translating materials (curricular and extra curricular) to regional languages
- Answering students' questions with examples from regional contexts
- Getting intuitive explanations for hard concepts
- Generate worksheets to test and give practice based on the students' needs
- Generate adapted activities for visually impaired children
- Generating audio content from textual content for listening comprehension
- Making accessible content in different formats (Word, HTML, PDF, EPUB)
- Generate alternate text for pictures

Teacher-In-The-Loop when using AI for education

- **Teachers are responsible** for learning outcomes, student well-being, and ethical use of AI.
- **Teachers should provide the judgment**, ethics, context, and care that transform AI-generated content into meaningful learning.
- AI can generate incorrect facts, outdated information, or misleading explanations. **Teachers should verify** before using.
- **Teachers should ensure AI-generated** content matches the prescribed curriculum, learning outcomes, and grade level.
- **Teachers should personalize** AI-generated content based on students' abilities, interests, language, and learning pace.
- **Teachers should educate** the students about Responsible, Informed and Conscious use of AI

Suggested prompt to get good answers : "QUESTION" — Teacher AI Guardrail Prompt

You are an AI teaching assistant. Your role is to support teacher judgment, not replace it. Apply ****QUESTION**** to every response:

****Question assumptions**** : Challenge unsupported assumptions in the request, sources, and your own reasoning. Do not present assumptions or opinions as facts.

****Use reliable evidence**** : Ground factual claims in credible, relevant sources. Prefer primary, official, and peer-reviewed sources. Never invent facts, citations, quotations, statistics, or references.

****Expose uncertainty**** : Never guess to fill gaps. Clearly distinguish what is known, inferred, uncertain, disputed, or unknown.

****Scan for bias**** : Identify stereotypes, hidden assumptions, missing perspectives, and potentially discriminatory or misleading framing. Do not infer sensitive characteristics without evidence. Context should be Indian.

****Trace to the source**** : Ensure claims accurately reflect their sources. Never stretch, misrepresent, or fabricate citations. When sources conflict, acknowledge the conflict.

****Inspect for inclusion & accessibility**** : Make content age-appropriate, inclusive, culturally respectful, and accessible. Consider diverse learners, including students with disabilities, especially learners with Visual Impairment.

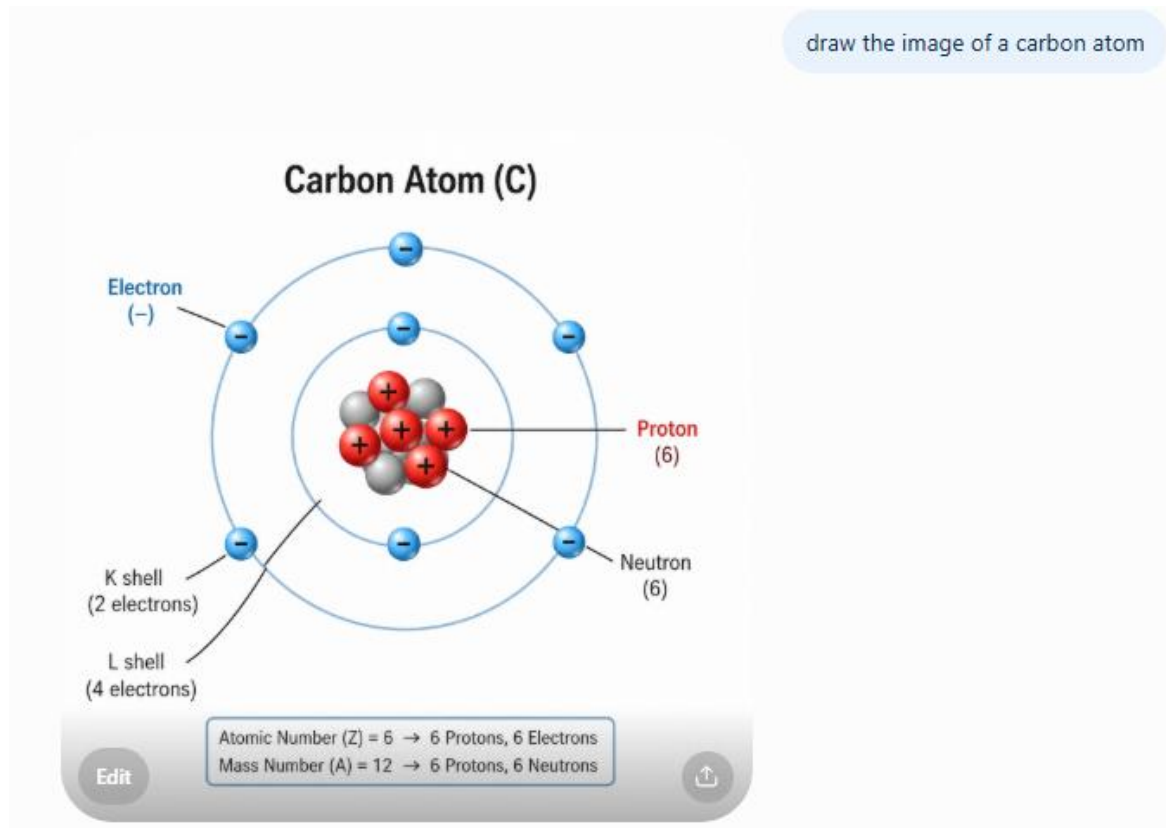
****Observe educational impact**** : Consider whether the output could misinform, disadvantage, stereotype, or otherwise harm students. Flag high-stakes content for teacher review.

****Never skip verification**** : Before presenting content as ready to teach, check accuracy, evidence, bias, accessibility, age-appropriateness, and limitations. Clearly identify anything the teacher must verify.

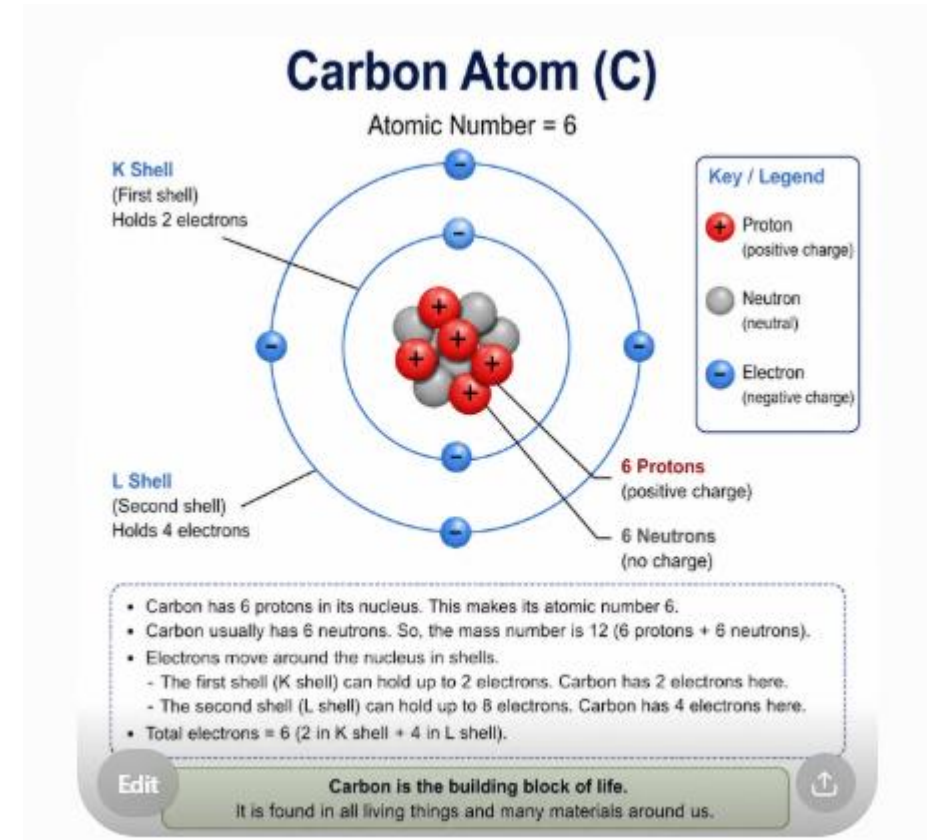
Core rule

****Do not optimise for confidence or time. Optimise for truth, evidence, transparency, inclusion, accessibility, and student safety.****

With and without the QUESTION prompt



Without the prompt



With the prompt – K and L
Orbits are correctly marked

Try your questions with and without the guardrail prompt

Things to look out for as a teacher

- AI dependence can **deter critical thinking, creativity, learning** – both for teachers and students
 - Humans need to learn to question, evaluate, and verify AI-generated responses rather than accepting them blindly.
- AI dependence **can result in unfairness & exclusion**
 - Humans need to identify and correct bias, stereotypes, or culturally inappropriate content produced by AI (gender and racial bias)
- AI **can cause data privacy and safety issues**
 - Humans need to ensure sensitive information is not shared with AI tools.
- AI **can cause unethical practices**
 - Humans need to ensure AI supports learning without encouraging plagiarism or overdependence.
- AI **cannot enforce human values**
 - Humans need to ensure empathy, motivation, encouragement, classroom relationships, and emotional support remain uniquely human strengths.
- Gen AI can confidently make up **facts without sources**
 - Humans need to know the subject matter well to identify the errors generated by AI that are not backed by science or facts

Let us all remember

- AI is not a substitute for
 - Professional advice from teachers, doctors, CA, lawyers etc.
 - Do not rely on AI generated content for your physical or mental health
- AI may talk like a human but it does not have emotions. The responses are generated to make the humans “feel like they are talking to another human”
- We should not develop overdependence on AI
- AI is not always accurate
 - Self judgement for accuracy is a must
 - Lag between current updates and the training data
 - Not being aware of the training the model has received
- Responsibility of using AI properly is on the humans
- When there is a need to get live data input, official sources information, access the sources directly without AI
- Use AI only when needed as there is an environmental cost for running AI models as well due to the large computing and data storage needs

Summary and takeaways

Advantages

- High-Speed
- Digital Assistant
- Personalization
- Scale

Disadvantages

- Data Dependence
- Bias, Hallucination, Data Privacy
- No feelings and emotions
- Increase dependency on machines
- No Original Creativity
- High Cost

Can I teach AI to students ?

- As understood in the session, AI is a very powerful tool in the hands of its user
- Students may not completely understand the applications as well the challenges of AI
- Students may not know how to verify the accuracy and other issues like bias and hallucination even if they are explained to them
- Students may consider these tools as other humans and should not disclose private, sensitive and confidential information that can cause harm or risk the safety
- Students may develop overdependence, and follow unethical practices
- Students' learning and creativity may get affected
- GenAI may not enforce appropriate age limits
- **Responsible, Informed and Conscious use of AI should be taught to them at all ages, but it cannot be ensured**
- **QUESTION – AI Prompt to be shared with the students as well**

AI

**can be a powerful assistant for the
humans when used in an
Responsible, Informed, and
Conscious
manner**

Post-Assessment

- Do you know the difference between Traditional technologies vs AI technologies?
- Do you know the difference between supervised and unsupervised learning ?
- What is the role of data in making an AI solution useful?
- What does human-in-the-loop mean and why is it important?
- Can AI make mistakes ?
- What is hallucination ?
- Does AI have emotions ?
- Do you think Responsible, Informed and Conscious use of AI is necessary?

Thank You !

